

Erik Hill

AI Evaluation & Testing Engineer

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SUMMARY

Self-taught AI evaluation and testing engineer building deterministic, replayable systems for measuring coding agents and LLM applications.

Career-changing from professional kitchens, I built a **verifiable-evaluation suite** whose claims are checked rather than asserted: **eleven published evaluation claims are independently replayed in CI**, deterministic, no LLM judges, failures published by design. Also **shipped a game to Google Play; 30+ public repos, two PyPI packages, one npm package**.

SKILLS

Agentic / AI: Multi-agent orchestration · LangGraph · Model Context Protocol (MCP) · LLM application development · Tool-calling & guardrails · RAG · Evals & regression gating (nDCG) · Prompt & context engineering · Safety (OWASP LLM Top 10, human-in-the-loop) · Differential-oracle, property-based & mutation testing · No-LLM-judge grading · Pre-registration & minimum-detectable-effect analysis

Backend & data: Python · FastAPI · Pydantic · SQLAlchemy 2.0 · PostgreSQL · pgvector · Alembic · REST / OpenAPI

Frontend / mobile: TypeScript · JavaScript · React · Next.js 15 · Java · Android (adb) · WebAssembly

Ship & verify: pytest · Vitest · JUnit · jqwik · Playwright · PyPI / npm packaging · GitHub Actions (matrix CI) · Docker · Cloudflare Pages (Wrangler) · Neon

SELECTED PROJECTS, all public at github.com/egnar09

- **agent-certlab:** Two real coding agents certified, every verdict regradeable from artifacts. A certification lab built on three **seeded-defect task families** with provable answer keys (incl. **coordinated two-file defects** no single-file patch can pass), a **precondition prover** (clean passes, every seed fails, it refused one of my own defects), and **calibration agents** the harness must separate before certifying.

- **reference-fleet:** An answer key for eval suites: six models each broken **one documented way at an exact seeded rate**, so detection rates are measurable against ground truth, a naive suite detects **1 of 6** (paired protocol). Plus a **LoRA-trained native defect**: mixture 0.507 → realized **0.200**: training's rate control is lossy, measured.

- **vac-protocol:** **Verifiable Agent Claims**: an evidence-bundle format + **offline verifier**; a registry where **eleven real claims are hash-checked and re-earned in CI**, with a **public record of it rejecting my own evidence**. Plus **vac-gate**: replayable evidence or no green check.

- **evalmut:** **Mutation testing for eval graders** (PyPI, with its **gradecore** engine): inject known defects, report which checks stayed green. Found **4 holes in its own engine's suite, 8 in ports of a widely-used framework's assertions. No LLM judge**; every operator tied to a documented real failure. **319 tests**.

- **rag-eval-lab:** deterministic RAG **eval harness** that catches a planted hallucination in CI; from-scratch **BM25 matching the published SciFact baseline (0.664 vs 0.665)**. **45 tests**.

- **model-drift:** public board, **frozen hash-fingerprinted suite across 17 models / 5 labs**, with a derived **minimum-detectable-regression floor**.

REVIEWED BY OTHERS

- **External audit of vac-protocol:** an outside engineer falsified my own verifier, cheapest forgery **four bytes**. Fix merged, then my instruments found the same defect class **four more times** (arXiv, below).

- **Peer review with Bartłomiej Nawara, PhD**, senior AI/ML engineer: his missing-not-at-random critique of my rate-limit handling reshaped **model-drift's reliability floor**, credited in the repo. I found **two design faults** in his pre-registered study, both adopted.

- **Merged compiler fix to TeaVM** (upstream) · **two open PRs to Promptfoo**.

PUBLICATIONS

“Four Ways to Forge a Bundle My Own Verifier Calls Clean: Refusal-Site Mutation Testing of an Evidence-Bundle Verifier.” Erik Hill. **arXiv preprint, 2026**. [arXiv:2608.26183](https://arxiv.org/abs/2608.26183)

Drift Notes: 21 articles on LLM evaluation failures, **85 comments from 25 readers**. erikhill.dev/notes · dev.to/agentdev9

EXPERIENCE

Independent AI Evaluation & Testing Engineer. Solo Practice · Remote · Mar 2026 – Present

- Designed a five-stage agent loop (Strategy → Execution → Critic → Evaluation → Ops) that plans, builds, reviews, and validates changes to a real React / Capacitor product, human-in-the-loop on every consequential step.
- Built per-role model routing and hook-based automation that hands work between stages with no manual relaunch; enforced rules with **property-based invariants** (jqwik) over a native engine and **adb-driven on-device validation**. Case study: github.com/egnar09/agentdev-harness

Indie Game Developer. SeraphLight Studios · Remote · Nov 2025 – Present

- **Shipped Tap Dodge Rush to Google Play** (Kotlin / Java, Android Views + Canvas, AdMob), closed testing with 12+ testers, then store review, then launch. Building a second, larger game (React / Capacitor), the product the harness above operates on.

Kitchen Manager / Shift Lead. Big Island Brewhaus & other high-volume kitchens · Hawaii · 2018 – 2025

- Led shifts and trained staff; owned prep, inventory, and food-safety under sustained pressure.

EDUCATION

Clark College, 2007–2009. · **freeCodeCamp**: self-directed software curriculum.